Causal Impact of Weather

on

Bike Demand

[1] Fanaee-T, Hadi, and Gama, Joao, "Event labeling combining ensemble detectors and background knowledge", Progress in Artificial Intelligence (2013): pp. 1-15, Springer Berlin Heidelberg, doi:10.1007/s13748-013-0040-3.

Executive Summary

Bad weather reduces bike demand by ~39%, not ~60% as naïve estimates suggest.

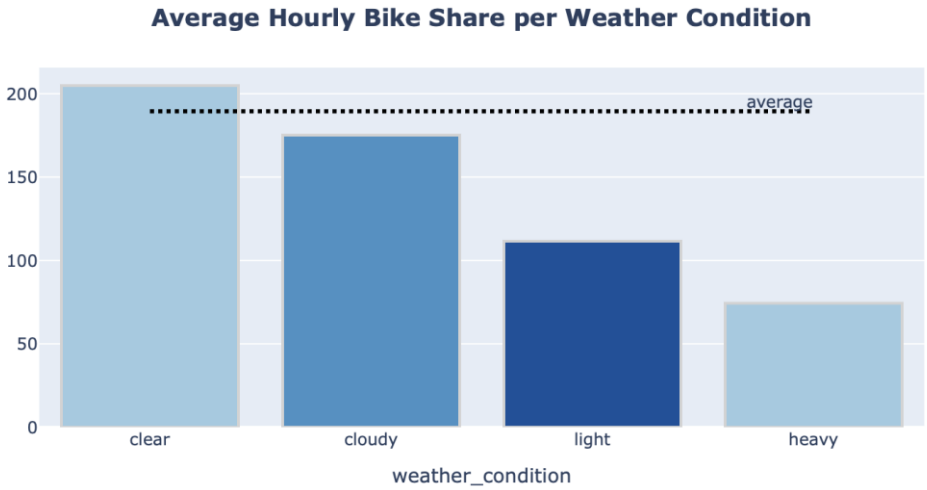
The effect varies significantly by user type and time of day:
- Casual users: −51% vs Registered: −36%
- Early morning: −29% vs Off-peak: −46%

Implication: **static strategies are suboptimal**.

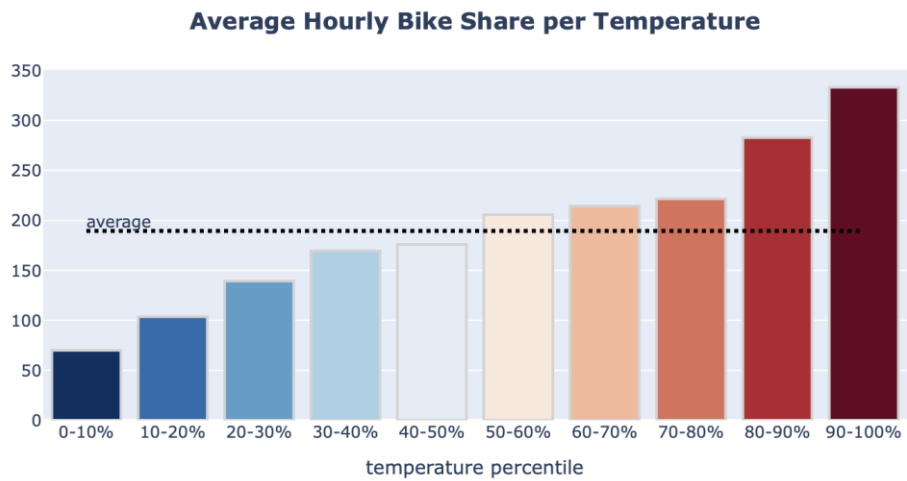
→ Target casual users for demand recovery during bad weather
→ Adjust supply and pricing dynamically by time of day

This would:
- Make operations more efficient; and
- Better capture demand under poor weather conditions

Problem & Causal Challenge



Light and Heavy rain significantly decreases demand



Extreme cold temperatures (0-20% percentile) influence demand the most

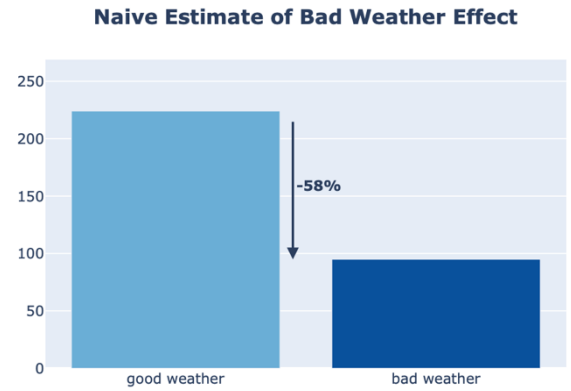
Defining bad weather

Light to heavy falls (rain, snow, ice)
and
Cold temperature (0-20% percentiles)

=> binary metric

- Bad weather impacts the demand of bike share, given it's riskier and less comfortable to cycle during poor weather conditions.
- Understanding the effect of bad weather on demand could lead to **optimizing business strategy** (e.g. decrease cost of supply rebalancing)

Why a naïve estimate is misleading



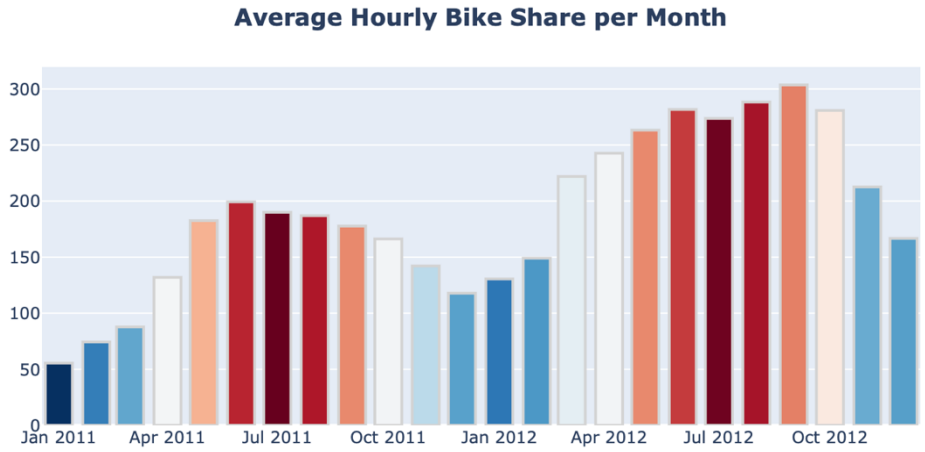
Naïve estimate: bad weather reduces demand by ~60%

This **estimate is biased** because weather is not randomly assigned.

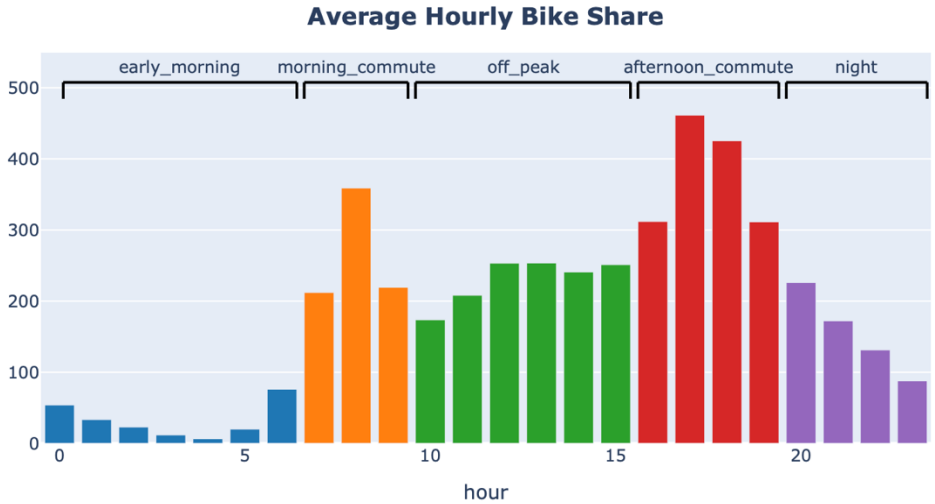
We should compare demand under different weather conditions within the same time context (hour, weekday, season) → **isolate the causal effect of weather**

Naïve method overestimates the effect.

Confounding Factors



Seasonality: winter → lower demand AND worse weather
and
User base is growing over time → higher demand in 2012 vs 2011



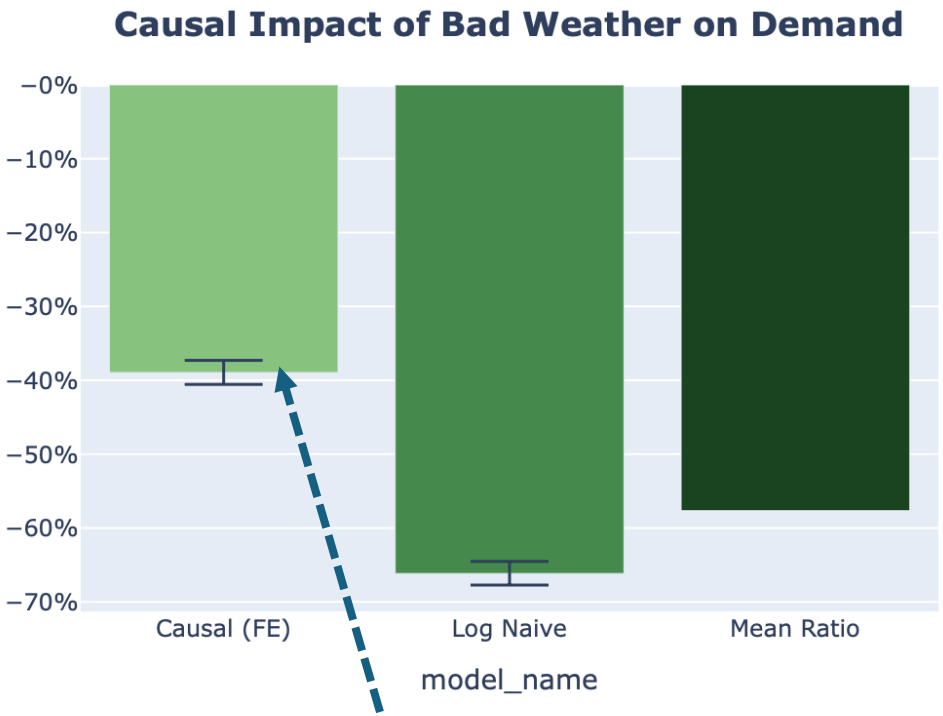
Time of day: night → lower demand AND colder temperatures

Estimating the causal impact of bad weather on demand

- Method: **regression model with time fixed effects** to control for confounding factors.
- Causal Estimate: **the causal effect of bad weather on bike demand is estimated at ~40%** (-39% ± 2% at 95% confidence)
- This compares demand under different weather conditions within the same hour of day, month, and year

Models

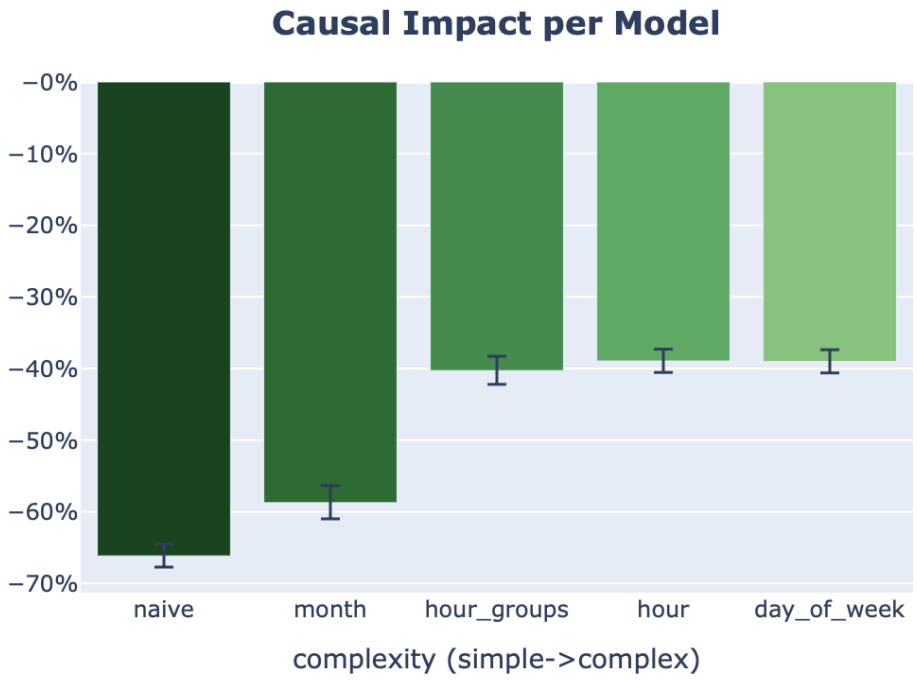
```
Causal (FE) Model:
     log(demand) ~ bad_weather + hour_of_day FE + year_month FE

Log Naïve:
    log(demand) ~ bad_weather

Mean Ratio:
    avg(demand in bad weather) / avg(demand in good weather) - 1
```



The causal effect is ~40%

As expected, naïve estimates overestimates the effect by overlooking confounding factors

Robustness of causal model



The estimated causal effect rapidly converges with model complexity to estimated ~40%

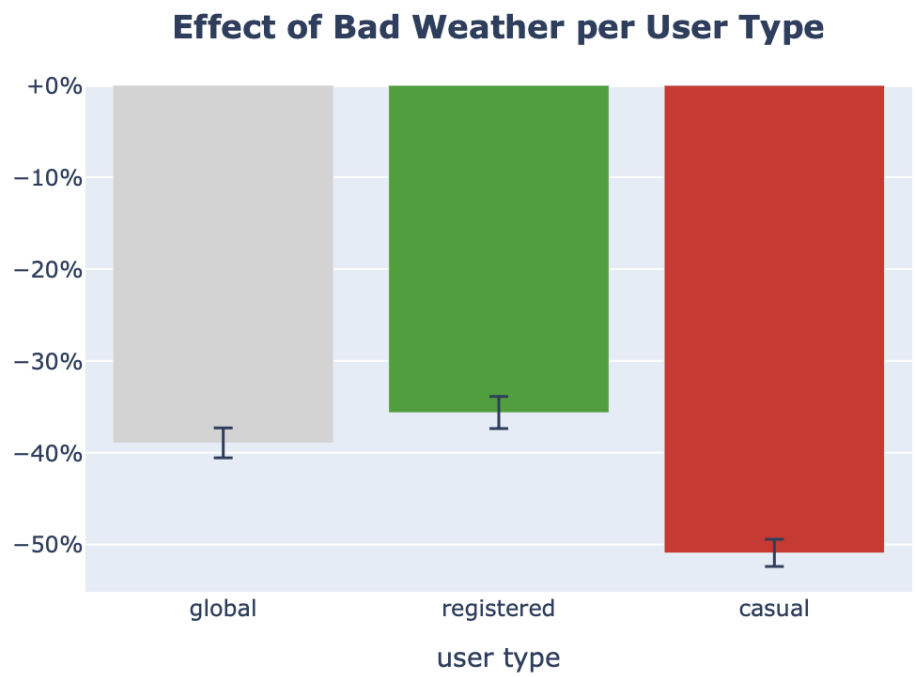
→ **effect stabilizes once time confounders are controlled**

→ **high confidence in estimate**

Models

```
naïve: log(demand) ~ bad_weather
month: log(demand) ~ bad_weather + year_month (FE)
hour_groups: log(demand) ~ bad_weather + year_month (FE) + hour_group (FE)
hour: log(demand) ~ bad_weather + year_month (FE) + hour (FE)
day_of_week: log(demand) ~ bad_weather + year_month (FE) + hour (FE) + day_of_week (FE)
```

Targeting casual users during bad weather



Bad weather reduces demand by ~39% overall

Casual users: −51% (highly elastic)
Registered users: −36% (more inelastic)
Difference: ~15% between casual and registered

→ **Casual usage is more discretionary**; while
→ **Registered usage is more driven by necessity**

Recommendation 1

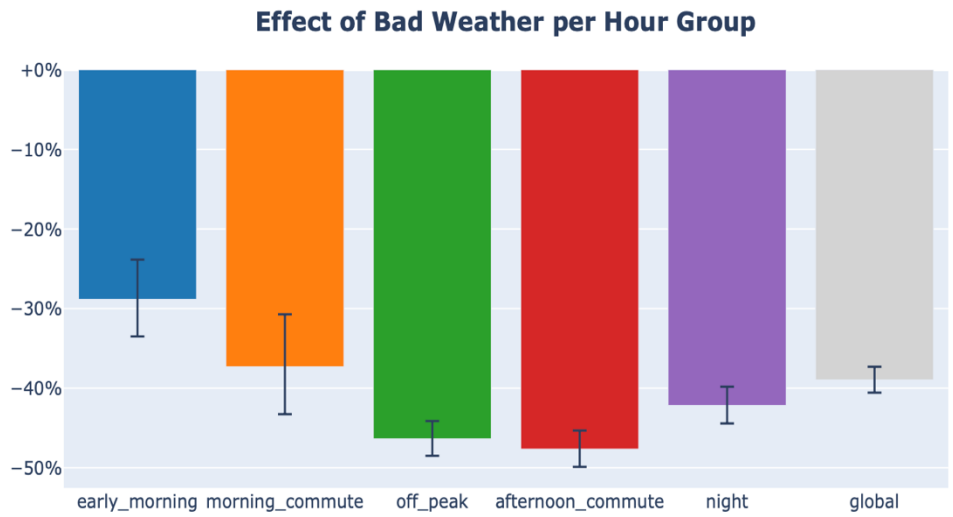
Target Casual Users during Bad Weather

Prioritize demand recovery initiatives on casual users:

- **Promotions / discounts during bad weather**
- **Targeted messaging (availability, convenience)**

→ Largest opportunity to recover lost demand

Demand Elasticity is time dependent



Demand sensitivity varies significantly by time:

- Most resilient: early morning (−29%)
- Most elastic: off-peak (−46%), afternoon commute (−48%)
- ~20% difference

Recommendation 2

Adjust supply and rebalancing dynamically

Reduce supply (in bad weather):

- off_peak (−46%)
- afternoon commute (−48%)

Maintain supply (in bad weather):

- early_morning (−29%)
- morning_commute (−37%)

→ Avoid over-supplying low-demand periods (cost inefficiency)
→ Avoid under-supplying high-value periods (lost rides, poor UX)

Uniform rebalancing is suboptimal

Limitations & Next Steps

Limitation 1 – Unobserved demand shocks

Events may affect demand independently of weather and introduce bias

Examples: concerts, strikes, transport disruptions

Limitation 2 – Geo-localization

Weather and demand vary across locations, but the model uses aggregated data, potentially hiding local effects

Limitation 3 – Simplified weather definition

Binary “bad weather” does not capture intensity or coupling effects

Potential Improvements

- Incorporate event and transport data
- Use location-level modeling
- Model weather as continuous + interactions

Key Takeaways

Bad weather reduces demand by ~39% (not 60%)

Effects are heterogeneous (user type, time of day)

Targeted interventions outperform uniform policies